Akhil Meda

TS Interim Clearance

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Education

Georgia Institute of Technology **Aug. 2025 – Present**

B.S. Computer Engineering — Expected Grad. Fall 2027 *Atlanta, Georgia*

Certifications

- CompTIA A+ (Feb 2024)
- Security+ (Apr 2024)
- Network+ (In Progress)

Skills

FPGA & Digital Design: combinational & sequential logic, FSMs, parameterized RTL, synthesizable SystemVerilog, streaming/backpressure concepts, verification testbenches (directed + randomized), scoreboards/checkers, waveform debug, latency/throughput optimization concepts
Networking / Telecom Foundations: TCP/IP & UDP fundamentals, Ethernet/packet-format basics, request/response debugging, packet analysis with Wireshark, latency/jitter measurement concepts
Programming: C, C++, SystemVerilog, Python, Java, SQL, JavaScript, PowerShell, RISC-V Assembly, x86 Assembly
Tools: Cadence Xcelium/SimVision, Git, Docker, VMware, Wireshark, Ghidra, n8n
Embedded: Linux interfaces (UART/SPI/I^2C concepts), hardware/software integration, basic digital I/O
Security: incident response, malware analysis, forensics, endpoint security, risk management, reverse engineering

Experience

Qualcomm **Summer 2026**

Incoming Network Engineer Intern

Alpha Team Solutions **Jan 2025 – Present**

Network & Cybersecurity Intern *Remote*

- Designed and deployed a secure GCP VPC with private subnets, routing, and firewall rules to isolate services and enforce least-privilege network access.
- Enabled and analyzed VPC Flow Logs to monitor traffic patterns, investigate connectivity issues, and support audit-ready network visibility.
- Owned end-to-end REST API and webhook integrations to trigger internal workflows reliably; implemented request validation, structured error handling, and operational safeguards for consistent execution.
- Instrumented requests and workflow runs with latency/throughput telemetry, trace identifiers, and audit logs; used Wireshark and system logs to diagnose timeouts, payload issues, and network-related failures.

Projects

FPGA for Trading Systems (Trading Club) *SystemVerilog, RTL Design, Verification* **Jan 2026 – Present**

- Led a team of 5 to build an FPGA-oriented "trade logging + booking" datapath: real-time capture and confirmation of high-rate transactions for replay, compliance, and latency-aware performance analysis.
- Designed 3 synthesizable SystemVerilog modules for streaming ingestion and control (FSMs, backpressure/flow-control handling, parameterized widths), targeting deterministic behavior and timing-friendly structure.
- Developing self-checking verification (directed + randomized vectors), reference-model comparison, and automated pass/fail to validate ordering, boundary conditions, and burst behavior.

Silicon Jackets: Parameterized Adder + Design Verification *SystemVerilog, Cadence Xcelium* **Jan 2026 – Present**

- Designing a parameterized adder RTL block with synthesizable SystemVerilog (configurable bit-width), emphasizing clean combinational logic and timing-friendly structure.
- Building reusable, self-checking DV in Cadence Xcelium with waveform-driven debug in SimVision; using directed vectors + randomized/regression runs with scoreboarding to isolate 1-bit mismatches.
- Adding functional checks (assertion-style properties where applicable) to validate carry behavior, overflow conditions, and boundary values across regressions.

GTXR Rocket Club: Custom PCB Driver Bring-Up *Hardware Integration, Embedded Systems* **Sep 2025 – Present**

- Verified low-level C drivers to support custom PCB boards, translating hardware register maps and electrical behavior into reliable software interfaces for flight/ground subsystems.
- Implementing peripheral bring-up and comms routines (UART/SPI/I^2C-style interactions, GPIO control, interrupts/timeouts), with careful handling of edge cases and recoverable failure modes.
- Validating end-to-end hardware/software integration through bench testing and iterative debug (signal verification, interface sanity checks, and firmware-level diagnostics) to de-risk system integration.